LEARNING TO IGNORE WHAT MATTERS: A JEPA WORLD MODEL KEEPS THE CUBE AND LOSES THE WALL

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ABSTRACT

We study a joint-embedding predictive world model (JEPA) trained with LeWM's objective, squared next-latent prediction plus a Gaussian marginal regularizer, no decoder and no reward, in a world where survival depends on one thing: a moving wall whose gap is revealed late. On the same data and ten-minute budget a generative flow model imagines the wall and commits to a gap; the grid JEPA imagines an empty room: the wall's latent signal is below the predictor's one-step residual, decoded rollouts lose the wall within two steps in all three model seeds, and termination heads barely read wall deaths. PPO with an MLP torso passes 0.27 walls per episode after training inside the JEPA (27 runs), against 0.94 after training in the real gate on the JEPA's own frozen encoders (nine runs) and 0.91 on true pixels; eight of nine runs inside the flow model pass 0.72–1.18 walls, the ninth 3.95. The real-gate comparison changes dynamics, reward and termination at once: it shows a deficit of the imagination pipeline, predictor and fitted heads together, and does not isolate the predictor. The wall the pipeline loses is decodable: a convolutional policy trained on the frozen latent decoded back to pixels learns the retreat-and-wait strategy in two of nine runs, as it does on true pixels in two of three. Sampling the fitted residual predictor does not improve control, and amplifying the wall's latent signal for the MLP policy produces no discovery. We prove that the objective allows this: a non-collapsed global optimum can carry no state at all, and an exactly Gaussian, fully decodable latent can hold a control-relevant feature at arbitrarily small amplitude. Non-collapse does not guarantee control-relevant state, and decodable state does not guarantee that imagination uses it.

1 INTRODUCTION

A world model is only as useful as the state it keeps. Joint-embedding predictive architectures (JEPAs) learn that state without a decoder: an encoder maps frames to latents, a predictor maps latent history and actions to the next latent, and both are trained on the squared prediction error plus a term that keeps the latent marginal Gaussian (LeCun, 2022; Assran et al., 2023; Balestriero & LeCun, 2025; Maes et al., 2026). Nothing forces such a model to render what it does not need. Nothing forces it to keep what it does need either. We build a world in which one feature decides survival and measure what a JEPA does with it.

The world is a gate (Figure 1B). A wall scrolls toward a cube that the agent moves; touching the wall or the frame border ends the episode with reward -1. The gap is drawn only once the wall is past $x = 40$: from the centre the cube sees it two steps before contact and needs five moves to reach it, so the strategy that survives is to retreat, wait for the reveal, then move. The world has a predictable part (the cube, the wall's approach) and one unpredictable bit (the gap side).

We train a grid JEPA with LeWM's objective (Maes et al., 2026) and a latent flow-matching model (Lipman et al., 2023; Hafner et al., 2025) on the same 500,000 random-action transitions under the same ten-minute budget, use each as a simulator for a recurrent PPO agent, and grade the agent in the real gate (Figure 1). The JEPA imagines a cube in an empty room, and none of its 27 agents (three model seeds, three predictor variants, three PPO seeds) ends training above 3.1% survival.

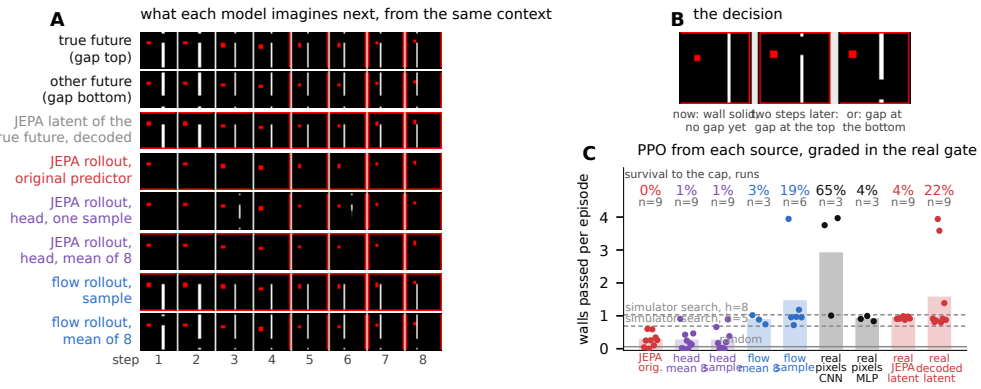


Figure 1: **One context, two possible futures, and the policies trained from each model.** A red cube must pass a white wall that scrolls left three pixels per step; the wall is solid until it passes $x = 40$, then a gap opens at the top or at the bottom with equal probability (B). **A**: eight imagined steps from one pre-reveal context (model seed 0, context 0 of Figure 2). The true future and its counterfactual differ only in the gap. The grid JEPA's latent of the true future decodes to the wall (gray label), but its rollout deletes the wall at step one and keeps the cube; a residual predictor on the frozen latent, sampled or averaged over eight, deletes it too. The flow model keeps the wall and commits to one gap, whether it samples or averages. **C**: PPO trained inside each model or in the real gate (pixels with convolutional or MLP torso, the frozen JEPA latent, or that latent decoded to pixels), graded in the real gate: walls passed per episode (bars: mean over runs, dots: runs; survival to the cap and the number of runs above; dashed: simulator search at horizons 5 and 8).

The flow model imagines the wall with one gap or the other; its agents pass at least 0.72 walls per episode in every run and find the retreat-and-wait strategy in 1 of 9, as PPO in the real environment does in 2 of 3.

The obvious suspect is the mean. For a fixed encoder the unrestricted squared-error optimum is the conditional mean (Lemma 1), the mean of two futures is not a future, and planning through a mean can incur maximal regret (Proposition 2). We tested the suspect inside each model, with a residual predictor fitted on the frozen JEPA latent and with the flow model, each used as one sample and as the mean of eight. Inside the JEPA the three predictors are indistinguishable: 0.27, 0.26 and 0.26 walls per episode, at most 3.1% survival. Inside the flow model the mean of eight keeps the wall and passes 0.9 walls per episode, three times the JEPA's, as the sampled model does in the runs that do not find the strategy. Certainty-equivalent planning is a real failure mode; these tests neither implicate it nor rule out failures caused by averaging.

The deficit is in imagination, not in the encoder alone. For model seed 0 the wall's amplitude in the JEPA latent, the squared distance between the latents of the true future with and without the wall, grows from 0.04 to 0.12 over eight steps, while the predictor's own one-step residual is 0.15 at the first step and 0.25 at the last, and the one-step prediction sits 0.4 of the way from the wall-free latent to the true one along the wall's own direction. The predictor rolls the wall away (Figure 2). Heads fitted with the death labels read border deaths off the latent at AUC 0.96 and wall deaths at 0.65. Two real-gate controls show a deficit relative to training on real transitions. The same MLP torso design passes 0.94 walls per episode on the frozen JEPA latent (nine runs, three encoders) and 0.91 on true pixels, against 0.27 inside the JEPA; real-gate training also replaces the fitted heads by the true reward, so the deficit is the imagination pipeline's, not the predictor's alone. And the wall is decodable: a convolutional policy trained on the frozen latent decoded back to pixels learns the retreat-and-wait strategy in 2 of 9 runs, both on encoder 1, as it does on true pixels in 2 of 3; the MLP policy learns it in none of the real-gate controls, amplified wall signal included, and once in nine runs inside the flow model (Appendix E). Doubling and quadrupling the JEPA's training budget leaves its agents at 0–3% survival, and a hundred-fold sweep of the regularizer's weight on the per-episode-texture cube world leaves the cube position unreadable.

The theory says the objective is allowed to do this. A non-collapsed global optimum of the joint objective can carry no controlled state at all, and any controller reading it is a state-blind policy (Proposition 1); an encoder that preserves state distances at a uniform scale pays the innovation of the dynamics in prediction loss (Corollary 1); and an exactly Gaussian, fully decodable latent can

hold a control-relevant feature at arbitrarily small amplitude, which makes the error of averaging away a fresh unpredictable bit arbitrarily cheap (Proposition 3; proofs in the text and Appendix A).

Contributions. (i) A controlled world in which one late-revealed bit decides survival, with a simulator reference and a state-blind baseline. (ii) The measurement that a JEPA trained with the published objective keeps the cube and loses the wall: amplitude below the predictor residual, deleted rollouts, blind heads, agents that never learn, over three model seeds, with longer fits on model seed 0. (iii) The same-model test in which sampled and averaged transitions fail alike, and the real-gate controls: policies trained on the JEPA's frozen latent pass 0.94 walls and, through a decoder, find the strategy, while policies trained on its rollouts average 0.27. (iv) Three main-text propositions on representation and planning, with proofs, that show the objective permits the failure, and an exact-ranking boundary result in the appendix. We study one JEPA instance in small worlds at a small budget; the cube worlds and a Craftax transfer in the appendix are exploratory.

2 THE OBJECTIVE, THE WORLD, AND THE MODELS

The objective. LeWM (Maes et al., 2026) jointly trains a pixel encoder $z_t = f_\theta(x_t) \in \mathbb{R}^D$ and a causal predictor of the next latent from H latents and actions:

$$\mathcal{L}(\theta) \;=\; \frac{1}{H}\sum_t \big\| g_\theta(z_{t-H+1:t}, a_{t-H+1:t}) - z_{t+1} \big\|_2^2 \;+\; \lambda\,\mathrm{SIGReg}\big(\{z_t\}\big), \qquad \lambda = 0.09. \tag{1}$$

Gradients reach the encoder through predictions and targets, without a stop-gradient, teacher, decoder or reward term. SIGReg, the test of Balestriero & LeCun (2025), uses P random unit directions u_p, K knots $t_k \in [0, 3]$, trapezoidal weights ω_k and window $w(t) = e^{-t^2/2}$,

$$\mathrm{SIGReg} = \frac{1}{P}\sum_{p,k} \omega_k\, w(t_k)\Big[\big(\widehat{\mathbb{E}}\cos(t_k\langle z, u_p\rangle) - w(t_k)\big)^2 + \big(\widehat{\mathbb{E}}\sin(t_k\langle z, u_p\rangle)\big)^2\Big], \tag{2}$$

with $K = 17$ and $P = 1024$. It compares projected empirical characteristic functions to a standard normal, constraining only the *marginal law* of z.

The gate. A red 6×6 cube moves in 63×63 RGB frames. Four actions move it three pixels left, up, right or down. Contact with a red ring one pixel inside the frame or with the wall ends the episode at reward -1. Other rewards are zero; the cap is 128 steps. A white wall three pixels wide enters at $x = 60$ every 22 steps and moves left three pixels per step. Its 16-pixel gap occupies rows 4–19 or 42–57 with probability $1/2$ each. The wall stays solid until $x \le 40$. Either gap is five moves from the vertical centre; a cube at $x = 28$ sees the gap at wall position $x = 39$ and is hit two steps later, while retreat to $x \le 16$ leaves six steps after the reveal. Random play has mean length 30.4 and never reaches the cap. Training uses $500{,}000$ uniform-random transitions from 100 environments, evaluation $25{,}000$; Appendix B describes six cube worlds (three backgrounds, still or hidden wind).

Three models, one budget. All models in the base comparison train on the same data for ten minutes on one A100, compilation included; the budget and regularizer sweeps use fixed step counts. The *grid JEPA* is Equation 1 with a 2.6M tile-transformer encoder, $4 \times 4 \times 32$ latent (512 numbers), and 8.3M factored spatial-temporal predictor with eight-frame history (Figure 1). The *class-token JEPA* is LeWM's six-layer width-192 ViT with a 192-number class-token read-out and AdaLN autoregressive predictor (15.3M parameters); it also fails the gate (Appendix G). The *flow model* is a latent flow-matching dynamics model (Lipman et al., 2023; Chen et al., 2024; Hafner et al., 2025): a reconstruction-trained tokenizer to the same $4 \times 4 \times 32$ latent, a causal transformer with factored attention over a 32-frame window, per-frame noise levels, four function evaluations per imagined frame, and reward and termination heads at loss weight 0.1. After JEPA training we freeze the encoder and fit reward, termination and behaviour-value heads on $80{,}000$ latents, a pixel decoder for the figures and the decoded-latent control, and a *residual predictor*, a small flow-matching head on the residual $z_{t+1} - g_\theta(\cdot)$ given the original prediction, the current latent and the action; none touches the JEPA's weights (Appendix C). Within-model comparisons isolate sampling; cross-family ones change several components.

3 THEORY

We separate representation sufficiency from planning with a point prediction; these are population statements under explicit function-class assumptions and do not describe optimization trajectories.

3.1 A NON-COLLAPSED, CONTROL-INSUFFICIENT GLOBAL OPTIMUM

For a latent law on $\mathbb{R}^D$, define the population regularizer

$$\mathrm{SIGReg}_\infty(\mathcal{L}(z)) = \int_{\mathbb{S}^{D-1}} \int_0^3 w(t) \left[\big(\mathbb{E}\cos(t\langle z, u\rangle) - w(t)\big)^2 + \big(\mathbb{E}\sin(t\langle z, u\rangle)\big)^2 \right] dt\, d\sigma(u), \quad (3)$$

where $w(t) = e^{-t^2/2}$ and σ is uniform on the sphere. This nonnegative functional vanishes for $z \sim \mathcal{N}(0, I_D)$; we use only this direction, and the finite-sample regularizer differs from it by an estimator bias (Appendix A).

Assume **(A1)** observations have form $x_t = R(p_t, b)$, and a measurable map β recovers the episode-constant nuisance b. Under the training distribution, b is independent of the controlled trajectory and actions. For deployment, assume b is independent of the initial controlled state and all exogenous transition randomness. Controlled dynamics and rewards do not otherwise depend on b. Assume **(A2)** a measurable map φ satisfies $\varphi(b) \sim \mathcal{N}(0, I_D)$. Assume **(A3)** the encoder class contains $\varphi \circ \beta$, and the predictor class contains the last-latent copy map. These realizability assumptions are substantive; finite pixel textures need not satisfy them exactly.

Proposition 1 (Texture trap)**.** *Under (A1)–(A3), an encoder and predictor attain zero population prediction loss and zero* SIGReg_∞*. Their latent has covariance* I_D *and satisfies* $I(z_t; p_{1:T}) = 0$ *under the training distribution. Any deployed controller observing only these latents, time, and its own actions has the value of a randomized state-blind policy, at most the supremum over such policies.*

Proof. Set $f^\star = \varphi \circ \beta$ and let $g^\star$ copy its last latent. Then $z_t = \varphi(b) = z_{t+1}$, so every prediction residual vanishes. The Gaussian marginal gives covariance I_D and zero population regularizer. Both objective terms are nonnegative, so their sum is globally minimized at zero. Training-distribution independence gives $I(\varphi(b); p_{1:T}) = 0$.

For deployment, draw the nuisance B and the controller's random seed U at reset, independently of the initial state and of the transition randomness; run the original controller in the environment with nuisance B, and let the state-blind policy, with private seed (B, U), feed the constant history $\varphi(B)$ to a copy of that controller and reproduce its actions with U. Coupled to the same initial state and transition randomness, the two policies choose the same first action, and if their actions agree through step t their states, latent histories and seeds agree, so their next actions agree; by induction trajectories, returns and terminations coincide. Nothing here assumes independence between b and states influenced by its induced actions. □

Corollary 1 (An innovation cost for state-resolving encoders)**.** *Let* $p_{t+1} = p_t + \mu(a_t) + w_t$ *and* $\mathcal{H}_t = \sigma(p_{\le t}, b, a_{\le t})$*. Write* $\Sigma_t = \mathrm{Cov}(w_t \mid \mathcal{H}_t)$ *and assume finite second moments for states and encoded observations. Assume observations have the same form* $x_t = R(p_t, b)$ *with fixed nuisance* b*. Suppose* $\|f(R(p, b)) - f(R(p', b))\| \ge \kappa \|p - p'\|$ *for all admissible* p, p', b*, with* $\kappa > 0$*. For every predictor using latent and action history,* $\mathbb{E}\|g(z_{\le t}, a_{\le t}) - f(x_{t+1})\|^2 \ge \kappa^2 \mathbb{E}\mathrm{tr}\Sigma_t$ *(Equation 5 in Appendix A). If* $\mathbb{E}\mathrm{tr}\Sigma_t > 0$*, this exceeds the texture encoder's zero prediction loss under (A1)–(A3).*

The bound is the innovation after conditioning on state history and nuisance, so a persistent wind is not automatically innovation, and it covers encoders that preserve every state distance at a uniform scale; Appendix A proves it without a conditional-independence assumption on the wind.

3.2 PLANNING WITH A MEAN

Lemma 1 (Squared prediction loss targets a conditional mean)**.** *Let* $\mathbb{E}\|z'\|^2 < \infty$*. For a fixed encoder and data law, over all measurable square-integrable predictors of* z' *from the latent history* c *and action* a*, the unique squared-error minimizer is* $m(c, a) = \mathbb{E}[z' \mid c, a]$*, up to null sets, and for every goal* $z_\star$

$$\mathbb{E}[\|z' - z_\star\|^2 \mid c, a] = \|m(c, a) - z_\star\|^2 + \mathrm{trCov}(z' \mid c, a). \quad (4)$$

The proof is the orthogonal decomposition in Appendix A. The second term is what a goal-image planner with a mean predictor drops; it is action-independent only in special cases (Proposition 4).

Proposition 2 (Certainty-equivalent planning can incur maximal regret)**.** *There exists a one-step problem with two actions and reward range* $[-1, 0]$ *where exact conditional-mean planning incurs regret* 1*. Here the planner chooses* $a_{\mathrm{CE}} \in \arg\max_a r(\mathbb{E}[y \mid a])$.

Proof. Let $S = [-2, -1] \cup [1, 2]$, with $r(y) = 0$ on S and $r(y) = -1$ elsewhere. Action A produces $y = \mp 3/2$ and action B produces $y = 0$ or 3, each outcome with probability $1/2$: $\mathbb{E}[r(y) \mid A] = 0$ and $\mathbb{E}[r(y) \mid B] = -1$, the means are 0 and $3/2$, the scores $r(0) = -1$ and $r(3/2) = 0$, so the mean planner selects B with regret 1, the whole reward range. Both outcome variances equal $9/4$. □

An exact expected-reward head solves this one-step problem without sampling; the proposition concerns the specified score, not every controller with a deterministic predictor. The boundary is classical: under affine dynamics with additive noise whose joint law is the same for every open-loop action sequence, and quadratic costs, an open-loop sequence costs its mean rollout plus a sequence-independent constant, so mean rollouts rank sequences exactly (Simon, 1956; Theil, 1957) (Proposition 4, Appendix A). Survival with termination and a $\{-1, 0\}$ reward is not an instance, and for nonlinear dynamics iterated one-step means need not be multi-step means.

3.3 Decodable information can have arbitrarily small amplitude

Proposition 3 (Small amplitude under an exactly Gaussian marginal)**.** *For every* $\varepsilon > 0$ *and integer* $D \geq 1$*, there is an observation process* $x_t = R(p_t, b)$ *with episode-constant nuisance* b*. Its state* $p_t \in \{0, 1\}$ *is a fresh fair bit, drawn independently of* b*, previous states, and previous actions. There is a measurable encoder* f_ε *with* $z_t \sim \mathcal{N}(0, I_D)$ *from which* p_t *and* b *are exactly recoverable almost surely. For the same nuisance, the two state encodings satisfy* $0 < \|f_\varepsilon(R(1, b)) - f_\varepsilon(R(0, b))\| < 2\varepsilon$*. Over all square-integrable measurable predictors of latent and action history through* t*, the optimal next-latent prediction is their midpoint* $m(b)$*, with* $0 < \mathcal{L}_{\mathrm{pred}}(f_\varepsilon, m) \leq \varepsilon^2$*,* $I(m(b); p_{t+1}) = 0$ *and* $\mathrm{SIGReg}_\infty(\mathcal{L}(z_t)) = 0$*. The state is control-relevant: with actions* $a_t \in \{0, 1\}$ *and reward* $\mathbf{1}\{a_t = p_t\}$*, latent feedback earns* 1 *per step and every state-blind policy* $1/2$*.*

Proof idea. Cut $\mathbb{R}$ into intervals of width 2ε and let the nuisance pick an interval with its Gaussian mass and a uniform position inside it; let the fair bit decide whether that position lands in the lower or the upper half of the interval's conditional Gaussian mass. The first latent coordinate is then exactly $\mathcal{N}(0, 1)$, the bit is read off from the half, the two encodings of the bit lie in one interval, and the optimal predictor is their midpoint, which depends on the nuisance only and costs at most ε^2. Appendix A gives the construction in full. □

The construction is measurable, with no finite-network or robust-decoding guarantee; it averages a fresh innovation and does not by itself explain the deletion of a predictable wall.

4 Controllers and protocol

PPO inside a model. A recurrent actor-critic (Schulman et al., 2017; Espeholt et al., 2018; Farebrother et al., 2024) is trained entirely on imagined rollouts: the model is reset from real eight-frame contexts, the policy acts on the model's latent, the model's reward and termination heads supply the return, and the trained policy is graded on 256 real episodes from reset. The policy class, horizon, optimiser and the number of imagined steps (1.2M) are identical across models. For the grid JEPA the transition is its original deterministic predictor, or the residual predictor used as one sample or as the mean of eight samples; for the flow model it is one sample or the mean of eight. The mean of eight has conditional covariance $\Sigma/8$: a mean-like transition, not the exact mean.

Planners. Goal-image MPC, LeWM's controller, minimises the predicted latent distance to a wall-free centre image; heads-MPC maximises the imagined return through the fitted heads; both search all 4^h action sequences ($h = 5$ on the gate) and replan every step. "Simulator search" is the same search on the true simulator with the true reward, no terminal value, and the unrevealed gap resampled per imagined copy.

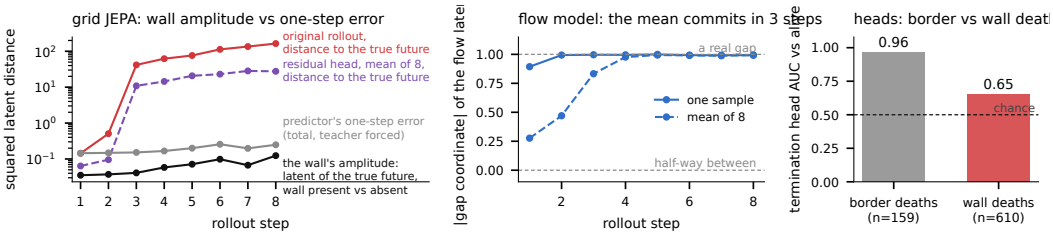


Figure 2: **Where the wall goes.** Left: squared latent distances in the grid JEPA (model seed 0) over 64 pre-reveal contexts; the wall's amplitude (latent of the true future with the wall against without it) lies below the predictor's one-step residual at every step, and the rollouts drift far from the true future. Middle: the flow model's latent on the axis between the two gap futures (40 contexts in which both branches survive); a sample commits at once, the mean of eight starts half-way and approaches a branch in three steps. Right: termination head fitted on the frozen JEPA latent, AUC of each death cause against alive transitions.

References. Random actions; a centre controller that reads the true state; a state-blind policy that alternates left and right; PPO in the real environment for 1.5M steps (Table 1 lists its observations and torsos); and the simulator search. Planning runs use 128 episodes with seven warm-up steps, policy runs 256 from reset, with shared random keys across arms (Appendix D); we report survival to the cap and walls passed per episode. Training seeds are the unit of replication: three JEPA model seeds with three PPO seeds per model and predictor variant; on the one flow checkpoint, six PPO seeds sampled and three for the mean of eight.

5 Results

5.1 The wall is decodable but lost in prediction

The wall's signal in the JEPA latent is smaller than the predictor's error (Figure 2, left). From 64 contexts in which the wall is solid at $x \in \{42, 45\}$ the true simulator renders three futures under the recorded actions: the recorded one, its counterfactual with the other gap, and the same trajectory with the wall removed. The encoder separates the true future from the wall-free one by a squared distance of 0.035 at step one and 0.12 at step eight, in a 512-dimensional latent of about unit variance per coordinate. The predictor's own teacher-forced one-step residual is 0.145 at step one and 0.25 at step eight. Write d for the difference between the latents of the true future with and without the wall, and the wall coordinate of a prediction for its projection on d, 0 at the wall-free latent and 1 at the true one: the teacher-forced one-step prediction has mean wall coordinate 0.39 at step one and 0.21 at step eight. The deterministic rollout is as close to the wall-free future as to the true one at every step, and its decoded frames contain no wall at any step, while the decoded latent of the true future contains the full wall (intensity 1.00 on the wall pixels; Table 12). Averaging cannot explain this: the wall is dropped before the gap is drawn, and the two possible futures share the wall. These are model seed 0; seeds 1 and 2 repeat it with wall amplitude 0.03 and 0.09 against a residual of 0.13 and 0.16 at step one, wall coordinate 0.45 and 0.66, termination heads at AUC 0.65 and 0.76 on wall deaths and agents at 0.17 and 0.33 walls; seed 2 keeps 0.68 of the wall's decoded intensity at step one and at most 0.02 after (Table 10).

The flow model keeps the wall in both modes. In its latent the same distance is 344 to 554; on the 40 contexts in which both branches survive the recorded actions, a sample sits at $|\text{gap coordinate}| = 0.89$ at step one and 0.99–1.00 after, the mean of eight at 0.28, 0.47, 0.83 and 0.98 (Figure 2, middle; Table 12). Its decoded frames show one gap or the other, never a half-open wall: a latent average need not decode to a pixel average.

The heads fitted on the latent see border deaths and barely see wall deaths (Figure 2, right; Table 13). Fitted on 80,000 real latents with the death labels, the termination head separates the 159 border deaths from alive transitions at AUC 0.963 and the 610 wall deaths at 0.649 (model seed 0; 0.88–0.96 and 0.65–0.76 over seeds); the reward head predicts -0.89 on border deaths and -0.05 on wall deaths, against -1 at every death. These are the heads the planners and in-model agents use, not a bound on any read-out.

Table 1: **Gate controllers, graded in the real environment.** Walls per episode and survival to the cap, mean ± s.d. over runs; "discovered" counts runs with final survival above 50%. Simulator search uses the true reward, no terminal value; real-gate PPO uses 1.5M steps.

controller	model seeds	policies	walls passed	survival (%)	discovered
grid JEPA, original predictor	3	9	0.27±0.22	0±1	0/9
grid JEPA, residual head, mean of 8	3	9	0.26±0.30	1±1	0/9
grid JEPA, residual head, one sample	3	9	0.26±0.32	1±1	0/9
flow model, mean of 8 samples	1	3	0.88±0.14	3±1	0/3
flow model, one sample	1	6	1.45±1.23	19±39	1/6
simulator, exhaustive search $h = 5$ (no terminal value)	–	–	0.69	1	–
simulator, exhaustive search $h = 8$ (no terminal value)	–	–	1.03	1	–
random	–	–	0.06	0	–
state-blind alternating	–	–	0.02	0	–
centre controller (true state)	–	–	0.00	0	–
PPO in the real gate, pixels, CNN torso	–	3	2.91±1.65	65±52	2/3
PPO in the real gate, pixels, MLP torso (latent policy class)	–	3	0.91±0.08	4±2	0/3
PPO in the real gate, frozen JEPA latent, MLP torso	3	9	0.94±0.04	4±1	0/9
PPO in the real gate, decoded JEPA latent, CNN torso	3	9	1.57±1.26	22±37	2/9

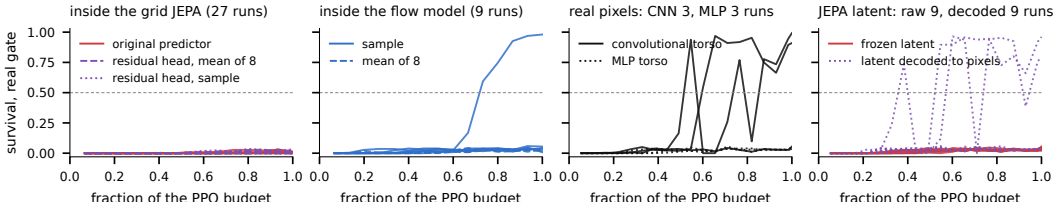


Figure 3: **Survival in the real gate over policy training**, one line per run; discovery is final survival above 50% (dashed line). Inside the grid JEPA no run leaves the floor. Inside the flow model and on true pixels with the convolutional torso some runs discover the retreat-and-wait strategy (dotted: the MLP torso on pixels, which does not in three runs). On the frozen JEPA latent no run discovers it; on the latent decoded back to pixels with the convolutional torso (dotted), 2 of 9 do. Walls passed, the quantity of the main comparison, are in Figure 1C and Table 1.

5.2 Nothing trained inside the JEPA learns to survive

Table 1 and Figure 3 give the control result. PPO inside the grid JEPA ends at or below 3.1% survival (8 of 256 episodes) at the final checkpoint in every one of its 27 runs: 0.27 ± 0.22 walls per episode with the original predictor, 0.26 ± 0.30 with the residual predictor as the mean of eight and 0.26 ± 0.32 as one sample (Table 9 lists every cell). The best cells, 0.9 walls at 3% survival, sit between horizon-5 and horizon-8 simulator search, and no run leaves the floor of Figure 3. Inside the flow model the eight runs without discovery pass 0.72 to 1.18 walls per episode, and one of the six sampled runs discovers the retreat-and-wait strategy and reaches 98% survival and 3.95 walls; PPO in the real environment discovers it in 2 of 3 runs (91, 5.5 and 99%). Discovery is a lottery in both. Discovery counts are 1 of 9 for the flow model and 0 of 27 for the JEPA (Fisher $p = 0.25$); all nine flow runs exceed the JEPA's mean of 0.27 walls and six exceed its best run (Mann–Whitney $p < 10^{-4}$). These run-level tests are conditional on the fitted models, and none of the 27 JEPA policies discovers the strategy.

Two controls in the real gate, with the true dynamics and the true reward, show a deficit relative to training on real transitions. PPO on the frozen JEPA latent (Figure 3, right) ends at 0.88–0.99 walls per episode and 2.3–5.5% survival in nine runs over three encoders, 0.94 against 0.27 inside the JEPA, and never the strategy. The same MLP torso design passes 0.84–1.00 walls on the flattened true frame in three runs, so the one-wall result is not specific to the JEPA latent. In the real gate the convolutional policy discovers the strategy in 2 of 3 pixel runs and, fed the frozen latent decoded back to pixels by the off-budget decoder, in 2 of 9; both decoded successes use encoder 1 (3.6–3.9 walls, 79–96%; 0.80–1.39 walls and 2–7% in the other seven), so the frozen latent supports discovery on at least one encoder. None of 42 transformed-latent runs discovers it; they pass 0.50–1.01 walls and test wall-signal gains of 4 and 16, four-fold scaling of probe, wall-difference and matched random subspaces, and decoded frames for the MLP torso (Appendix E). The comparison that carries our claim is within one policy class, 0.94 walls on the frozen latent in the real gate against 0.27 inside the JEPA; real-gate training also replaces the fitted heads by the true reward and

changes the collection protocol (Appendix C), so it measures the imagination pipeline, predictor and heads together, and does not separate transition error from head error.

Two references bound what the planners can say. Simulator search with the wall known, the gap hidden until the reveal and no terminal value passes 0.69 walls at horizon 5 and 1.03 at horizon 8; revealing the gap to horizon-5 search moves it only to 0.70: the horizon binds, not the gap. LeWM's goal-image and heads planners on the JEPA pass 0.04 to 0.16 walls (Appendix G); in a horizon-3 audit the heads-MPC's selected return, -0.46, is a maximum over 64 noisy plans with mean -0.84 (Table 13).

5.3 Sampling the fitted residual predictor does not rescue control

The comparison is inside one model. In the grid JEPA the residual predictor has lower held-out one-step error than the original (1.3×10^{-4} per coordinate as the mean of eight, 2.1×10^{-4} as one sample, against 3.1×10^{-4}) and is a distribution rather than a point; neither moves the agent: 0.26 and 0.26 walls against 0.27, the same 0–3% survival (Table 1). No distributional predictor is ruled out; this one does not help. In the flow model the mean of eight passes 0.88 ± 0.14 walls in three runs, against 0.96 ± 0.16 for the five sampled runs without discovery and 3.95 for the one with it; 0 against 1 discoveries in three against six runs says nothing about the mode. Proposition 2 remains true and Figure 2 (middle) shows the half-way latent it warns about, but the averaged transition approaches a branch within three steps (gap coordinate 0.83) and keeps the wall; inside the JEPA it is gone in either mode. These tests neither identify the mean as the cause nor rule out failures caused by averaging.

5.4 Budget, regularizer, and the cube worlds

Training budget. Training the grid JEPA for twice and four times the steps ($17{,}000$ and $34{,}000$) lowers the prediction loss from 0.0006 to 0.0003 and leaves the heads' held-out reward correlation at 0.31–0.33 and PPO inside the model at 0–3% survival in all three predictor variants (model seed 0, PPO seed 0; Table 11).

Regularizer dose. On the cube world with a random texture fixed within each episode, where the class-token JEPA behaves like the texture optimum of Proposition 1, sweeping the SIGReg weight over 0, 0.009, 0.09 and 0.9 and the latent width over 32 and 192 (five arms) does not recover the cube position: the held-out reward correlation of the fitted heads stays between -0.01 and 0.00 and the decoder's error at 0.0028–0.0029 in every regularized arm; at weight 0 the latent collapses (decoder error 0.055; Table 4). The weight sets how Gaussian the latent is, not what it is a function of.

Cube worlds and a state-blind caveat. The six cube worlds (Appendix B) show the two representation failures the theory predicts: with a per-episode random background the class-token JEPA reaches near-zero prediction loss by encoding the background and its reward-driven controllers fall to 0–2%; with a hidden wind the grid JEPA stays in the noise basin. A policy that alternates left and right without looking survives the still cube worlds in 100% of episodes (Table 3); still-world survival is not evidence.

Craftax, exploratory. A grid JEPA trained on Craftax-Classic pixels reaches six-action goal positions in 94.5% of trials against 18.8% for random plans, while its reward head's correlation with the true reward falls from 0.92 on training transitions to 0.35 held out and reward-MPC returns 0.47 over 100 steps against 1.45 for random actions (Appendix D): what it keeps steers the agent and does not say what is worth reaching.

6 Related work

Joint-embedding prediction learns its targets in representation space (LeCun, 2022; Assran et al., 2023; Bardes et al., 2024); LeWM trains encoder and action-conditioned predictor jointly under squared loss and SIGReg (Maes et al., 2026; Balestriero & LeCun, 2025); other systems plan on frozen features (Assran et al., 2025; Zhou et al., 2024; Sobal et al., 2025). Anti-collapse methods constrain the learning dynamics or the latent statistics (Grill et al., 2020; Chen & He, 2021; Zbontar

et al., 2021; Bardes et al., 2022; Jing et al., 2022); our counterexample keeps full marginal rank and loses control-relevant information.

Sobal et al. (2022) showed that joint-embedding objectives prefer a fixed background to a moving object while tolerating distractors redrawn each frame; Tang et al. (2023) and Ni et al. (2024) analyse self-predictive learning and its state abstractions. We add the exactly Gaussian nuisance-only optimum of the population objective, the innovation cost of keeping state, the amplitude construction that separates decodability from weight in the loss, and diagnostics of counterfactual wall distances in a trained model. Control-oriented alternatives define relevance through reward, transitions or value (Ferns et al., 2004; Zhang et al., 2021; Grimm et al., 2020; Farahmand et al., 2017; Tishby et al., 2000), and Lambert et al. (2020) document the mismatch between prediction and control quality that our worlds instantiate.

Stochastic video prediction has a long history (Mathieu et al., 2016; Babaeizadeh et al., 2018; Denton & Fergus, 2018; Lee et al., 2018); recent world models keep stochastic predictions for imagination and control (Hafner et al., 2019; 2023; Alonso et al., 2024; Bruce et al., 2024; Hafner et al., 2025), our reference uses flow matching (Lipman et al., 2023; Chen et al., 2024), and policy learning inside a model follows Sutton (1991); Ha & Schmidhuber (2018); Hafner et al. (2020). Certainty equivalence and the value of information are classical (Simon, 1956; Theil, 1957; Howard, 1966; Bar-Shalom & Tse, 1974); Propositions 2 and 4 restate their consequences for latent planning; sampling-based MPC propagates model uncertainty (Chua et al., 2018).

7 Discussion and limitations

What the objective guarantees. Equation 1 asks the latent to be predictable and Gaussian. Predictability rewards features whose future is a function of the past; Corollary 1 prices the rest for encoders that keep state at a uniform scale, while the constructed nuisance encoder pays nothing. Gaussianity guarantees nothing: Proposition 1 has an exactly Gaussian latent with no state, and Proposition 3 an exactly Gaussian, fully decodable latent whose one useful bit has arbitrarily small amplitude. The gate JEPA shares with the second case the split between decodability and weight in the loss, and does not measure it: the wall is decodable, its amplitude is below the predictor's residual, and the predictor drops it; raising that amplitude for the policy produced no discovery. The constructions say the objective permits this, not why training lands there: the wall the model drops is predictable, not an innovation.

What we did not find. We expected the mean predictor to be the failure, and the theory still says it can be (Proposition 2; Proposition 4 gives one sufficient condition for exact open-loop ranking). Our tests do not identify it as the cause here: inside the JEPA, samples and eight-sample means give the same agents, and the flow model's mean keeps the wall and approaches a branch within three steps.

Remedies. The deficit sits in the imagination pipeline: policies trained on the frozen latent pass 0.94 walls and, through a decoder, sometimes find the strategy; policies trained on the JEPA's rollouts with its heads average 0.27 (Section 5.2). The flow model is a feasibility reference, not an isolated test: it keeps the wall in decoded frames and differs in tokenizer supervision, history and reward supervision at once. Sampling the residual predictor, longer gate fits, the regularizer sweep on the fixed-background cube world (Section 5.4) and an amplified wall signal for the policy produce no discovery. Which change would make the pipeline carry the wall is beyond this paper; the diagnostic targets for this system are not: rollouts whose decoded frames keep the wall, and a head that reads wall deaths.

Limitations. One JEPA objective, two encoders, small worlds, ten-minute budgets, a four-fold budget sweep; no published LeWM benchmark is reproduced, and a larger encoder or budget could keep the wall. The MLP torso does not find the strategy on true pixels in three runs either, so discovery is compared across torsos and walls passed within one class. Discovery is a lottery (2 of 3 on pixels): the flow model's 1 of 9 is a count, not a rate.

REPRODUCIBILITY STATEMENT

The worlds, models, controllers and every diagnostic are JAX programs; the base world-model comparison uses a ten-minute budget on one A100 including compilation, the fixed-step sweeps report their times, and the off-budget heads, decoders and policies have the budgets given in Appendix C. Section 4 states the evaluation protocol; Appendix D the metrics. All numbers in the text are produced by one script from the run ledgers. Code, data generation and the run ledgers will be released as an anonymous artifact.

STATEMENT ON THE USE OF AI

Language models were used to draft and edit prose, to check the proofs for gaps, to write analysis and plotting scripts, and to act as an adversarial reviewer of the manuscript. Every experiment, number, proof and claim was specified, run and verified by the authors.

REFERENCES

Eloi Alonso, Adam Jelley, Vincent Micheli, Anssi Kanervisto, Amos Storkey, Tim Pearce, and François Fleuret. Diffusion for world modeling: Visual details matter in atari. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

Mahmoud Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, Mojtaba Komeili, Matthew Muckley, Ammar Rizvi, Claire Roberts, Koustuv Sinha, Artem Zholus, and Nicolas Ballas. V-JEPA 2: Self-supervised video models enable understanding, prediction and planning, 2025.

Mohammad Babaeizadeh, Chelsea Finn, Dumitru Erhan, Roy H. Campbell, and Sergey Levine. Stochastic variational video prediction. In *International Conference on Learning Representations (ICLR)*, 2018.

Randall Balestriero and Yann LeCun. LeJEPA: Provable and scalable self-supervised learning without the heuristics, 2025.

Yaakov Bar-Shalom and Edison Tse. Dual effect, certainty equivalence, and separation in stochastic control. *IEEE Transactions on Automatic Control*, 19(5):494–500, 1974.

Adrien Bardes, Jean Ponce, and Yann LeCun. VICReg: Variance-invariance-covariance regularization for self-supervised learning. In *International Conference on Learning Representations (ICLR)*, 2022.

Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Ballas. Revisiting feature prediction for learning visual representations from video, 2024.

Jake Bruce, Michael Dennis, Ashley Edwards, Jack Parker-Holder, Yuge Shi, Edward Hughes, Matthew Lai, Aditi Mavalankar, Richie Steigerwald, Chris Apps, Yusuf Aytar, Sarah Bechtle, Feryal Behbahani, Stephanie Chan, Nicolas Heess, Lucy Gonzalez, Simon Osindero, Sherjil Ozair, Scott Reed, Jingwei Zhang, Konrad Zolna, Jeff Clune, Nando de Freitas, Satinder Singh, and Tim Rocktäschel. Genie: Generative interactive environments. In *International Conference on Machine Learning (ICML)*, 2024.

Boyuan Chen, Diego Martí Monsó, Yilun Du, Max Simchowitz, Russ Tedrake, and Vincent Sitzmann. Diffusion forcing: Next-token prediction meets full-sequence diffusion. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Xinlei Chen and Kaiming He. Exploring simple siamese representation learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

Kurtland Chua, Roberto Calandra, Rowan McAllister, and Sergey Levine. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

Emily Denton and Rob Fergus. Stochastic video generation with a learned prior. In *International Conference on Machine Learning (ICML)*, 2018.

Lasse Espeholt, Hubert Soyer, Remi Munos, Karen Simonyan, Volodymyr Mnih, Tom Ward, Yotam Doron, Vlad Firoiu, Tim Harley, Iain Dunning, Shane Legg, and Koray Kavukcuoglu. IMPALA: Scalable distributed deep-RL with importance weighted actor-learner architectures. In *International Conference on Machine Learning (ICML)*, 2018.

Amir-massoud Farahmand, Andre Barreto, and Daniel Nikovski. Value-aware loss function for model-based reinforcement learning. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, pp. 1486–1494, 2017.

Jesse Farebrother, Jordi Orbay, Quan Vuong, Adrien Ali Taiga, Yevgen Chebotar, Ted Xiao, Alexander Irpan, Sergey Levine, Pablo Samuel Castro, Aleksandra Faust, Aviral Kumar, and Rishabh Agarwal. Stop regressing: Training value functions via classification for scalable deep reinforcement learning. In *International Conference on Machine Learning (ICML)*, 2024.

Norm Ferns, Prakash Panangaden, and Doina Precup. Metrics for finite markov decision processes. In *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2004.

Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre H. Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Daniel Guo, Mohammad Gheshlaghi Azar, Bilal Piot, Koray Kavukcuoglu, Rémi Munos, and Michal Valko. Bootstrap your own latent: A new approach to self-supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

Christopher Grimm, André Barreto, Satinder Singh, and David Silver. The value equivalence principle for model-based reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

David Ha and Jürgen Schmidhuber. World models, 2018.

Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *International Conference on Machine Learning (ICML)*, 2019.

Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. In *International Conference on Learning Representations (ICLR)*, 2020.

Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse domains through world models, 2023.

Danijar Hafner, Wilson Yan, and Timothy Lillicrap. Training agents inside of scalable world models, 2025.

Ronald A. Howard. Information value theory. *IEEE Transactions on Systems Science and Cybernetics*, 2(1):22–26, 1966.

Li Jing, Pascal Vincent, Yann LeCun, and Yuandong Tian. Understanding dimensional collapse in contrastive self-supervised learning. In *International Conference on Learning Representations (ICLR)*, 2022.

Nathan Lambert, Brandon Amos, Omry Yadan, and Roberto Calandra. Objective mismatch in model-based reinforcement learning. In *Conference on Learning for Dynamics and Control (L4DC)*, 2020.

Yann LeCun. A path towards autonomous machine intelligence, 2022. Version 0.9.2, OpenReview.

Alex X. Lee, Richard Zhang, Frederik Ebert, Pieter Abbeel, Chelsea Finn, and Sergey Levine. Stochastic adversarial video prediction, 2018.

Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023.

Lucas Maes, Quentin Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. LeWorldModel: Stable end-to-end joint-embedding predictive architecture from pixels, 2026.

Michael Mathieu, Camille Couprie, and Yann LeCun. Deep multi-scale video prediction beyond mean square error. In *International Conference on Learning Representations (ICLR)*, 2016.

Tianwei Ni, Benjamin Eysenbach, Erfan Seyedsalehi, Michel Ma, Clement Gehring, Aditya Mahajan, and Pierre-Luc Bacon. Bridging state and history representations: Understanding self-predictive RL. In *International Conference on Learning Representations (ICLR)*, 2024. URL `https://openreview.net/forum?id=ms0VgzSGF2`.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017.

Herbert A. Simon. Dynamic programming under uncertainty with a quadratic criterion function. *Econometrica*, 24(1):74–81, 1956.

Vlad Sobal, Jyothir S V, Siddhartha Jalagam, Nicolas Carion, Kyunghyun Cho, and Yann LeCun. Joint embedding predictive architectures focus on slow features, 2022. URL https://arxiv.org/abs/2211.10831.

Vlad Sobal, Wancong Zhang, Kyunghyun Cho, Randall Balestriero, Tim G. J. Rudner, and Yann LeCun. Learning from reward-free offline data: A case for planning with latent dynamics models, 2025.

Richard S. Sutton. Dyna, an integrated architecture for learning, planning, and reacting. *ACM SIGART Bulletin*, 2(4):160–163, 1991.

Yunhao Tang, Zhaohan Daniel Guo, Pierre Harvey Richemond, Bernardo Avila Pires, Yash Chandak, Remi Munos, Mark Rowland, Mohammad Gheshlaghi Azar, Charline Le Lan, Clare Lyle, András György, Shantanu Thakoor, Will Dabney, Bilal Piot, Daniele Calandriello, and Michal Valko. Understanding self-predictive learning for reinforcement learning. In *International Conference on Machine Learning (ICML)*, 2023. URL https://proceedings.mlr.press/v202/tang23d.html.

Henri Theil. A note on certainty equivalence in dynamic planning. *Econometrica*, 25(2):346–349, 1957.

Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method, 2000.

Yi Wan, Muhammad Zaheer, Adam White, Martha White, and Richard S. Sutton. Planning with expectation models. In *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 3649–3655, 2019.

Yi Wan, Ali Rahimi-Kalahroudi, Janarthanan Rajendran, Ida Momennejad, Sarath Chandar, and Harm van Seijen. Planning with expectation models for control, 2021.

Jure Zbontar, Li Jing, Ishan Misra, Yann LeCun, and Stéphane Deny. Barlow twins: Self-supervised learning via redundancy reduction. In *International Conference on Machine Learning (ICML)*, 2021.

Amy Zhang, Rowan McAllister, Roberto Calandra, Yarin Gal, and Sergey Levine. Learning invariant representations for reinforcement learning without reconstruction. In *International Conference on Learning Representations (ICLR)*, 2021.

Gaoyue Zhou, Hengkai Pan, Yann LeCun, and Lerrel Pinto. DINO-WM: World models on pre-trained visual features enable zero-shot planning, 2024.

A PROOFS

Propositions 1 and 2 are proved in Section 3; Figure 4 illustrates the one-step regret example.

Corollary 1.

Proof. Put $Z = f(x_{t+1})$ and $\mathcal{C}_t = \sigma(z_{\leq t}, a_{\leq t})$. Because observations and latents are measurable functions of state and nuisance, $\mathcal{C}_t \subseteq \mathcal{H}_t$. Any predictor using a shorter window is also $\mathcal{C}_t$-measurable. Conditional least squares and the tower property give

$$\begin{aligned}\mathbb{E}\|g - Z\|^2 &\geq \mathbb{E}\|Z - \mathbb{E}[Z \mid \mathcal{C}_t]\|^2 \\ &= \mathbb{E}\|Z - \mathbb{E}[Z \mid \mathcal{H}_t]\|^2 + \mathbb{E}\|\mathbb{E}[Z \mid \mathcal{H}_t] - \mathbb{E}[Z \mid \mathcal{C}_t]\|^2 \\ &\geq \mathbb{E}\mathrm{trCov}(Z \mid \mathcal{H}_t).\end{aligned}$$

The cross term vanishes because its first factor has conditional mean zero given $\mathcal{H}_t$. Take conditionally independent copies P, P' of p_{t+1} given $\mathcal{H}_t$. Such copies exist for Euclidean state spaces, using their regular conditional distributions. The nuisance b and current state p_t are fixed under this conditioning. The conditional independent-copy variance identity yields

$$\begin{aligned}\mathrm{trCov}(Z \mid \mathcal{H}_t) &= \tfrac{1}{2}\mathbb{E}[\|f(R(P,b)) - f(R(P',b))\|^2 \mid \mathcal{H}_t] \\ &\geq \tfrac{\kappa^2}{2}\mathbb{E}[\|P - P'\|^2 \mid \mathcal{H}_t] \\ &= \kappa^2 \mathrm{trCov}(p_{t+1} \mid \mathcal{H}_t) = \kappa^2 \mathrm{tr}\Sigma_t.\end{aligned}$$

The final equality follows because $p_t + \mu(a_t)$ is $\mathcal{H}_t$-measurable. Taking expectations proves Equation 5. Averaging these inequalities gives the corresponding bound for a multi-time prediction objective. Proposition 1 supplies the zero-loss comparator when its assumptions also hold. □

Lemma 1.

Proof. Let $m = \mathbb{E}[z' \mid c, a]$. For every square-integrable measurable $g(c, a)$, expand $g - z' = (g - m) + (m - z')$. The cross term has expectation zero since $\mathbb{E}[m - z' \mid c, a] = 0$. Hence

$$\mathbb{E}\|g - z'\|^2 = \mathbb{E}\|g - m\|^2 + \mathbb{E}\mathrm{trCov}(z' \mid c, a).$$

The second term does not depend on g. The first is nonnegative and vanishes exactly when $g = m$ almost surely. The same expansion conditional on (c, a), with $g = z_\star$, proves Equation 4. □

The innovation bound. Corollary 1 states

$$\mathbb{E}\|g(z_{\leq t}, a_{\leq t}) - f(x_{t+1})\|^2 \geq \kappa^2 \mathbb{E}\mathrm{tr}\Sigma_t. \tag{5}$$

The finite-sample regularizer. The finite-sample regularizer differs from Equation 3. For independent identically distributed samples with characteristic function ϕ and empirical estimate $\widehat{\phi}_n$,

$$\mathbb{E}|\widehat{\phi}_n(t) - w(t)|^2 = |\phi(t) - w(t)|^2 + \frac{1 - |\phi(t)|^2}{n}. \tag{6}$$

Indeed, $\mathbb{E}\widehat{\phi}_n = \phi$ and $\mathbb{E}|\widehat{\phi}_n - \phi|^2 = (1 - |\phi|^2)/n$, since each complex exponential has modulus one; expanding around ϕ gives the identity. At a Gaussian law, the residual equals $(1 - e^{-t^2})/n$. This is estimator bias, not an optimization floor. Changing the law changes both terms; dependent temporal samples introduce additional covariance terms. Finite projections and knots also do not characterize Gaussianity.

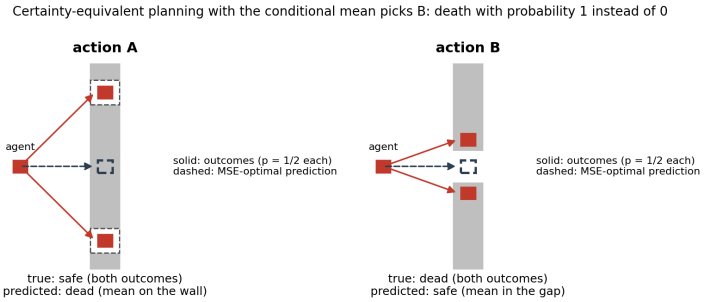


Figure 4: The one-step witness of Proposition 2. Safe pads are $S = [-2, -1] \cup [1, 2]$ and everything else is death. Action A always lands in a pad, but its mean is in the wall. Action B always dies, but its mean is in a pad. Solid arrows are the two equally likely outcomes, the dashed arrow is the squared-error-optimal prediction, and the predictor is correct about both means.

Exact ranking under affine dynamics and quadratic costs.

Proposition 4 (Exact open-loop ranking under affine dynamics and quadratic costs)**.** *Fix z_1 and deterministic matrices A, B, Q, with $Q \succeq 0$. For each deterministic open-loop action sequence, let $z_{t+1} = Az_t + Ba_t + w_t$. The joint noise law is identical across sequences, has finite second*

moments, and satisfies $\mathbb{E}w_t = 0$. *Temporal independence is not required. Define* $m_1 = z_1$, $m_{t+1} = Am_t + Ba_t$, *and* $J(a_{1:h}) = \mathbb{E}\sum_{t\le h}(z_t - z_\star)^\top Q(z_t - z_\star)$. *Then*

$$J(a_{1:h}) = \sum_{t\le h}(m_t - z_\star)^\top Q(m_t - z_\star) + \sum_{t\le h}\mathrm{tr}\big(Q\mathrm{Cov}(z_t)\big). \tag{7}$$

The second sum is action-independent, so mean rollouts rank all open-loop sequences exactly.

Expected features also suffice when value is linear in them (Wan et al., 2019; 2021).

Proof. Unrolling the two recursions gives

$$z_t = A^{t-1}z_1 + \sum_{k<t} A^{t-1-k}(Ba_k + w_k), \qquad m_t = A^{t-1}z_1 + \sum_{k<t} A^{t-1-k}Ba_k.$$

Thus $e_t = z_t - m_t = \sum_{k<t} A^{t-1-k}w_k$ has mean zero and finite second moments. Its law is the same for every deterministic action sequence by the joint-noise assumption. Consequently $\mathbb{E}z_t = m_t$ and $\mathrm{Cov}(z_t) = \mathbb{E}[e_t e_t^\top]$ is action-independent. For each t, expand

$$(z_t - z_\star)^\top Q(z_t - z_\star) = (m_t - z_\star)^\top Q(m_t - z_\star) + 2(m_t - z_\star)^\top Qe_t + e_t^\top Qe_t.$$

The cross term has expectation zero. The last term has expectation $\mathrm{tr}(Q\mathbb{E}[e_t e_t^\top]) = \mathrm{tr}(Q\mathrm{Cov}(z_t))$. Summing proves Equation 7. Its second sum is the same additive constant for every sequence, so pairwise rankings agree, including ties. Temporal correlations are already included in $\mathbb{E}[e_t e_t^\top]$; no temporal-independence assumption is needed. □

Proposition 3.

Proof. Partition $\mathbb{R}$ into $I_j = [2\varepsilon j, 2\varepsilon(j+1))$, $j \in \mathbb{Z}$. Let $q_j = \Pr(G \in I_j)$ for $G \sim \mathcal{N}(0,1)$. Every q_j is positive and their sum is one. Let F_j be the conditional Gaussian distribution function on I_j, with inverse quantile function Q_j on $(0,1)$. The conditional CDF is continuous and strictly increasing on the interval interior, so its inverse is measurable and takes values in that interior.

Draw J with probabilities q_j, $U \sim \mathrm{Uniform}(0,1)$, and $V \sim \mathcal{N}(0, I_{D-1})$, independently. Set $b = (J, U, V)$; for $D = 1$, omit V. Take $R(p,b) = (p,b)$ and define

$$f_\varepsilon(R(p,b)) = \big(Q_J((U+p)/2), V\big). \tag{8}$$

The mixture $(U + p_t)/2$ is uniform on $(0,1)$, since p_t is an independent fair bit. Conditional on $J = j$, the first coordinate therefore has the Gaussian law restricted to I_j. Mixing with weights q_j recovers $\mathcal{N}(0,1)$. Independence of V gives the joint law $\mathcal{N}(0, I_D)$ and zero population regularizer.

Given the first coordinate y, its interval determines J almost surely. Write $v = F_J(y)$. Then $p_t = \mathbf{1}\{v > 1/2\}$ and $U = 2v - p_t$ almost surely; the other coordinates recover V. Thus the latent recovers both state and nuisance. For fixed b, the two first coordinates are distinct and lie inside the same interval of width 2ε. All remaining coordinates agree, proving the distance bound.

Latent history reveals b, while the next bit remains independent and fair. The next latent consequently has two equiprobable values, say $z^{(0)}(b)$ and $z^{(1)}(b)$. Lemma 1 gives $m(b) = (z^{(0)}(b) + z^{(1)}(b))/2$. Its conditional squared error is $\|z^{(1)}(b) - z^{(0)}(b)\|^2/4$, strictly positive and bounded by ε^2. The midpoint is square-integrable by Jensen's inequality, since both state encodings have finite second moments. Taking expectations proves the loss bound, including strict positivity since the conditional error is positive almost surely. The midpoint depends only on b, so it is independent of p_{t+1}. Finally, decoding the current bit permits $a_t = p_t$. A state-blind action is independent of the current fair bit, giving success probability $1/2$. □

Remark 1 (Information must enter the rollout or its score). A planner that scores two candidate futures identically when they differ only in a future observation and the response to it assigns no value to that information; the restriction must hold for its rollout and its score, terminal value included. An exact closed-loop terminal value can favour retreat before a reveal even with an open-loop rollout, and replanning can exploit later observations (Howard, 1966; Bar-Shalom & Tse, 1974). Our oracle search tests a specified horizon and score; it does not bound all feedback controllers.

B THE CUBE WORLDS

The six cube worlds share the gate's body (frame, cube, actions, border ring, cap, reset distribution uniform on $[16, 42]^2$) without the wall and cross three backgrounds with two dynamics (Figure 5). The background is black, independent gray pixel noise redrawn every frame, or gray noise fixed throughout an episode. Dynamics are deterministic or include an unrendered wind added to each move; each wind coordinate follows an integer random walk with increments $-1, 0, +1$ at probabilities $0.1, 0.8, 0.1$, clipped to $[-2, 2]$ (mean absolute wind 0.83 pixels per axis). Fixed texture is predictable nuisance, fresh texture is unpredictable nuisance, wind changes the controlled motion. On these worlds the class-token JEPA behaves like the texture optimum of Proposition 1 when the background is fixed per episode (prediction loss 0.0004, cube position unreadable, Tables 5 and 6) and the grid JEPA stays in the noise basin under wind (prediction loss 0.73 with per-frame noise). Figure 5 shows the seven worlds and Figure 6 three real gate episodes. Table 2 gives the round-1 survival of every controller, Table 3 the edge-start re-grading of Section 5.4, and Table 4 the regularizer sweep on the fixed-background still world.

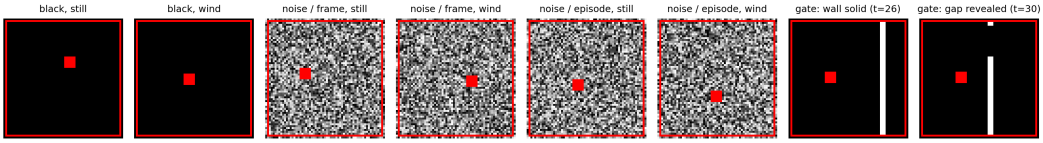


Figure 5: The seven worlds. Three backgrounds are crossed with deterministic motion or hidden wind; the wind is not rendered, so still and wind observations have matching appearance distributions. The gate is shown before and after its reveal.

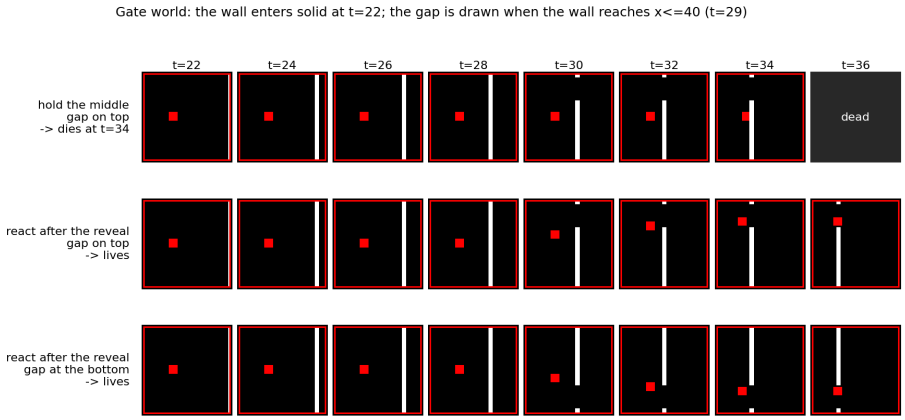


Figure 6: Three real gate episodes. The wall enters at $t = 22$ and reveals its gap at $t = 29$. The upper trajectory dies at $t = 34$; the lower trajectories retreat and then respond to the revealed gap. Before the reveal, the rendered wall does not identify its gap side.

Table 2: Survival to the 128-step cap in the six cube worlds, round-1 controllers. The original still-world reset distribution admits a safe alternating state-blind policy, so still-world survival does not require state feedback.

world	random	centre	LeWM goal	LeWM heads	LeWM PPO	grid goal	grid heads	grid PPO	flow PPO px	flow PPO z	PPO real
black / still	20	100	97	100	100	88	100	100	100	100	100
black / wind	0	99	64	60	99	1	5	94	94	96	98
noise per frame / still	20	100	5	9	0	92	100	100	98	99	100
noise per frame / wind	0	99	0	0	0	0	0	0	97	1	96
noise per episode / still	20	100	80	2	0	100	85	2	98	51	100
noise per episode / wind	0	99	0	0	0	77	8	0	95	1	78

C HYPERPARAMETERS AND COMPUTE

Each world supplies $500{,}000$ uniform-random transitions. Model training uses bfloat16 compute, float32 parameters, and one A100. A 500-step calibration estimates the training length fitting 575

Table 3: The same controllers re-graded from edge starts (cube reset 3–8 pixels from a border), survival in percent. Planners use 128 episodes and horizon 3; policies are the saved round-1 policies graded on 256 episodes.

world (edge starts)	random	alternate	centre	LeWM goal	LeWM heads	LeWM PPO	grid goal	grid heads	grid PPO	flow PPO px	flow PPO z	PPO real
black / still	8	100	100	69	93	91	41	99	99	100	97	99
black / wind	0	0	99	58	55	98	1	3	94	96	39	96
noise per frame / still	8	100	100	4	11	0	47	100	100	96	51	99
noise per frame / wind	0	0	99	0	0	0	0	0	0	94	0	93
noise per episode / still	8	100	100	50	5	0	89	75	2	93	25	99
noise per episode / wind	–	–	–	0	0	0	70	15	0	93	0	82

Table 4: SIGReg dose-response for the class-token JEPA on the fixed-background still cube world, each arm at the arm's calibrated 24,000 steps: final prediction and regularizer losses, the off-budget pixel decoder's MSE, the held-out reward-head correlation, and the survival of the two planners at horizon 3.

SIGReg weight	latent	pred. loss	SIGReg	decoder MSE	held-out reward r	goal MPC	heads MPC
0.0	192	0.0000	51.47	0.05541	-0.01	55	59
0.009	192	0.0001	2.14	0.00284	-0.00	69	4
0.09	192	0.0004	0.62	0.00278	-0.01	80	2
0.9	192	0.0088	0.57	0.00289	0.00	58	19
0.09	32	0.0004	0.60	0.00282	-0.01	76	2

seconds including compilation; the grid gate model trains for 8,500 steps (the budget sweep of Table 11 for 17,000 and 34,000), the class-token cube models for 23,750–24,000. Model seeds 0, 1, 2 of the grid JEPA differ only in the initialisation and data-order seed.

JEPA settings. Both arms use batch 128 and AdamW with learning rate and weight decay 10^{-3}, 10% warm-up followed by cosine decay to zero, and gradient clipping at 1.0. SIGReg uses weight 0.09, 17 knots on $[0, 3]$, and 1024 random projections. Neither arm uses a stop-gradient, teacher, decoder, or reward during world-model fitting.

	class-token JEPA (`lewm6`)	grid JEPA (`tile_factored`)
encoder	ViT, 7×7 patches	tile transformer + strided conv
width / depth / heads	192 / 6 / 3	128 / 6 / 8
encoder MLP	768	$4\times$ width
read-out	class token	4×4 grid, 32 channels
latent dimension	192	512
projector	linear-BN-GELU-linear, 2048	same, hidden 512
predictor	AdaLN-zero autoregressive	spatial then causal temporal
predictor depth / heads / head dim	6 / 16 / 64	6 / 8 / 32
predictor MLP / width	2048 / latent	$2\times$ width / 256
history	3 frames	8 frames
dropout	0.1	0
parameters	15.3M	2.6M + 8.3M

Flow settings. The tokenizer uses 9,000 steps, batch 256, and learning rate 9×10^{-4}; dynamics 16,000 steps, batch 64, the same learning rate, width 256, six layers, eight spatial and temporal attention heads, a 32-frame context, independent per-frame noise levels, a shortcut loss of weight 0.25 every fourth step after the first quarter of training, history dropout 0.25, two history frames resampled every eight steps after half of training, and a temporal loss weight of 0.25. The symlog reward head has 255 bins over $[-20, 20]$; reward and termination losses have weight 0.1 and enter the trunk at full strength for half of training, then at factor 0.1. Inference uses four function evaluations per frame. We corrected terminal-transition masking in the recipe's reward loss (Appendix H).

JEPA read-outs. Post-hoc heads use the frozen encoder and 80,000 frames from a different behaviour-policy seed, with the last 200 time rows held out. The pixel decoder uses 8,000 steps, batch 256, learning rate 10^{-3}, width 256, six residual blocks; it supports the figures and the decoded-latent control, in which its frozen output, clipped to $[0, 255]$, replaces the pixels of the real-gate PPO; no true pixel enters that policy after the encoder. Reward, termination, and behaviour-value heads are two-layer perceptrons of width 512 on the latent and the one-hot action, 6,000 steps, batch 1024, learning rate 10^{-3}, Monte Carlo value targets at discount 0.99, value loss weight 0.1. The residual predictor models $z_{t+1} - m$ with $m = g(z_{t-7:t}, a_{t-7:t})$ the original grid prediction, conditioned on

Table 5: Round-1 model diagnostics. JEPA columns report prediction loss and held-out reward-head correlation; flow columns report gFVD, action faithfulness, and termination AUC on real and imagined rollouts.

	LeWM (class token)		grid JEPA		flow model			
world	pred. loss	reward r	pred. loss	reward r	gFVD	AF	done AUC	im. done AUC
gate	0.0012	0.53	0.0006	0.33	0.0053	0.972	1.000	0.769
black / still	0.0012	0.22	0.0005	0.97	0.0000	1.000	0.905	0.916
black / wind	0.0644	0.89	0.0391	0.88	0.0005	0.825	0.999	0.694
noise per frame / still	0.7341	0.01	0.0007	0.98	0.1650	1.000	0.935	0.928
noise per frame / wind	0.7440	0.01	0.7345	-0.01	0.1516	0.973	0.998	0.630
noise per episode / still	0.0004	-0.01	0.0012	0.09	0.3427	1.000	0.920	0.878
noise per episode / wind	0.0004	-0.00	0.0031	0.33	0.3134	0.872	0.997	0.717

Table 6: Preliminary class-token diagnostics on the cube worlds. Effective rank is the exponential entropy of the normalized covariance eigenvalues out of 192; position read-outs (pixels of error) use ridge regression and two-layer perceptrons on standardized or whitened latents against the constant-mean guess.

World	eff. rank	ridge	MLP (std.)	MLP (whit.)	mean guess
`off_still`	47.7	19.56	18.53	0.71	19.56
`off_wind`	43.8	1.36	0.10	0.01	19.84
`gray_frame_still`	166.4	19.61	32.35	32.85	19.56
`gray_episode_still`	110.6	20.55	17.93	30.00	19.56
`gray_episode_wind`	122.7	20.64	20.84	24.17	19.84

m, z_t, and the action: three residual blocks of width 1024, 8,000 steps, batch 2048, 74 seconds, four function evaluations per sample. On 20,000 held-out windows the original predictor's squared error is 3.07×10^{-4}, one residual sample 2.05×10^{-4}, the mean of eight 1.32×10^{-4}.

Policy optimisation. PPO uses discount 0.99, GAE parameter 0.9, clipping 0.2, value coefficient 0.5, entropy coefficient 0.01, AdamW at 2.5×10^{-4} with weight decay 0.01, four epochs per batch, and gradient clipping at 1.0. Pixels pass through a three-block Impala torso, latents through a two-layer perceptron; a GRU receives that embedding and the previous action; the critic uses 64 categorical bins with histogram-Gaussian targets. Real-environment PPO uses 128 environments and 64 steps per iteration with four minibatch groups, truncated backpropagation over 16 steps, and 1.5M environment steps. In-model PPO uses 150 updates of 512 trajectories of 16 steps (1.2M imagined steps) with minibatch size 128; half of the starts are resets, half dataset states with their real eight-frame context. All reported policy rows use final checkpoints.

Compute. Logged training times on a shared A100: grid JEPA training 573 s (8,500 steps; 1,074 and 3,802 s for the 17,000- and 34,000-step fits); probe and heads 3 min; residual predictor 74 s; PPO inside the JEPA 2.0–4.3 min per run; PPO inside the flow model 5–6 min per run at four function evaluations, 20–26 min with the mean of eight; PPO in the real environment 3.7–7.7 min on pixels with the convolutional torso and 51–70 s with the MLP torso, 76–78 s on the frozen latent, 4.7–4.8 min on the decoded latent with the convolutional torso and 1.8–2.9 min with the MLP torso, 1.3–3.2 min on the transformed latents; simulator search at horizon 8: 7 s; the latent diagnostic of Figure 2: 2 min.

D Evaluation protocol

A run fixes a world, a model checkpoint, and a controller. Planning uses 128 episodes; policies 256. Episodes end on death or after 128 steps. Within each harness, start draws and per-step environment randomness are shared across controllers. Planning requires seven random warm-up steps to obtain an eight-frame history, and episodes dying during warm-up are rejected and resampled, which conditions the planning start distribution on survival; policy and baseline rows start at reset. The simulator search of Section 5.2 uses the planning harness and its random row is identical, episode for episode, to the planners' random row. Survival is the fraction alive at the cap; length is the mean number of steps from reset; walls passed counts walls leaving the frame while the agent is alive; deaths per 100

steps is 100 times deaths divided by total lived steps. Episode standard errors describe one trained system; training seeds are reported as separate runs.

The latent diagnostic. From the evaluation set we take 64 eight-frame contexts whose last frame has the wall solid at $x \in \{42, 45\}$ and reconstruct the simulator state; the recorded future, its counterfactual with the other gap, and the wall-free future are then rendered with the recorded actions. Amplitudes are squared distances between the encoder's latents of two rendered futures at the same step; the one-step residual is the predictor's teacher-forced error on the recorded future; the gap coordinate is the projection of a rollout latent on the unit vector from the counterfactual's latent to the recorded one, centred at their midpoint and scaled so that the two futures sit at ± 1. Decoded intensities are the mean minimum colour channel over the wall's pixels and over each 16-row gap band.

Craftax. A grid JEPA trained on Craftax-Classic pixels is graded by six-action goal reaching from 128 environments and by reward-MPC over 100 steps with horizon-six CEM (Table 7).

Table 7: Craftax-Classic results with the grid JEPA and post-hoc heads.

grid JEPA on Craftax-Classic	planned	random	oracle
closed loop: goal reached (fraction)	0.945	0.188	1.000
closed loop: position error (L_1)	0.08	1.53	0.00
reward MPC: random return	1.45		
reward MPC: myopic return	0.78		
reward MPC: greedy return	0.94		
reward MPC: plan return	0.47		
behaviour policy return	4.07		

E CONTROLS ON THE LATENT POLICY

Table 8 lists every real-gate PPO run whose observation was built from the frozen JEPA latent or chosen to match it. The 42 transformed-latent runs use model seeds 0 and 1 and three policy seeds each; with the reference arms (pixels with either torso, frozen and decoded latent with the convolutional torso over three model seeds) the table holds 66 runs, all at 1.5M steps. The latent policy is an MLP on the flattened observation, the pixel policy the Impala CNN (Appendix C); the MLP torsos on latents and on flattened frames share the design, not the input width. *Torso.* The MLP torso on the flattened true frame passes 0.84–1.00 walls and survives 2–6%, the level of the frozen latent, and does not discover the strategy in three runs; on the decoded latent it does the same in six. *Amplified wall signal.* With z_0 the encoder's output on the same frame with the wall pixels erased, the policy observes $z + (g-1)(z - z_0)$, the wall's own latent difference scaled g-fold (its squared amplitude g^2-fold); $g = 4$ and $g = 16$ produce no discovery. *Subspaces.* A ridge probe (ridge 1, centred, 80/20 split of 32,000 random-frame latents of model seed 0) maps the latent to one-hot wall position (22 classes) and gap state (undrawn, top, bottom). Held out, it scores 0.70 on wall position against a 0.71 majority rate and 0.90 on gap state against a 0.90 majority rate, and on the frames whose gap is drawn it names the side in 0.4% (0% for model seed 1): it answers "undrawn". Its 22-dimensional span holds 0.3% of the latent variance against 4.3% for a random span of the same rank. The wall-difference vectors $z - z_0$ of 8,192 frames need 306 of 512 directions for 90% of their energy (295 for model seed 1; top direction 1.8%); that span carries 90% of the difference energy and 80% of the latent variance, against 60% and 60% for a random span of the same rank. For a span with orthogonal projector P the policy observes $z + (g-1)Pz$ with $g = 4$; scaling the probe span, the wall-difference span, or a random span matched to either rank this way produces no discovery.

F ROUND-2 TABLES

G ROUND-1 GATE TABLE AND WALL MASS

Table 14 lists the round-1 gate controllers, one run each; Table 15 the thresholded wall-mass diagnostic that preceded the latent measurement of Section 5.1; Figure 7 the round-1 decoded futures.

Table 8: **Controls on the latent policy in the real gate.** Walls passed per episode and survival to the cap, mean $\pm$ s.d. over runs; "discovered" counts runs with final survival above 50%.

observation of the real-gate PPO	runs	walls passed	survival (%)	discovered
pixels, convolutional torso	3	2.91±1.65	65±52	2/3
pixels, MLP torso	3	0.91±0.08	4±2	0/3
frozen JEPA latent, MLP torso	9	0.94±0.04	4±1	0/9
decoded JEPA latent, convolutional torso	9	1.57±1.26	22±37	2/9
decoded JEPA latent, MLP torso	6	0.92±0.08	3±2	0/6
frozen latent, wall signal $\times 4$, MLP torso	6	0.89±0.08	3±1	0/6
frozen latent, wall signal $\times 16$, MLP torso	6	0.93±0.05	3±1	0/6
frozen latent, probe wall subspace $\times 4$	6	0.89±0.05	3±1	0/6
frozen latent, random subspace of rank 22 $\times 4$	6	0.85±0.12	3±1	0/6
frozen latent, wall-difference subspace $\times 4$	6	0.93±0.04	4±1	0/6
frozen latent, random subspace matched to the wall-difference rank $\times 4$	6	0.86±0.18	4±2	0/6

Table 9: PPO inside the grid JEPA for every model seed, predictor variant and PPO seed: survival in percent and walls passed.

		survival (%)			walls passed		
predictor	model seed	PPO 0	PPO 1	PPO 2	PPO 0	PPO 1	PPO 2
original predictor	0	0	0	1	0.04	0.25	0.60
	1	0	0	1	0.00	0.26	0.59
	2	0	1	0	0.10	0.34	0.27
residual head, mean of 8	0	3	0	0	0.91	0.02	0.00
	1	2	1	0	0.43	0.25	0.00
	2	0	0	1	0.17	0.13	0.46
residual head, one sample	0	2	1	0	0.66	0.17	0.00
	1	0	0	0	0.02	0.02	0.00
	2	0	3	1	0.20	0.89	0.38

H THE REWARD-HEAD CORRECTION IN THE FLOW RECIPE

The original recipe masks transitions entering reset frames out of reward training. Our worlds assign their only nonzero reward, -1, on terminal transitions, so masking those transitions removes the entire reward signal: the head predicts zero, reward correlation is undefined, and every policy has zero discounted return. The corrected head predicts r_{t+1} from the frame-t summary and action a_{t+1} with terminal transitions kept in its loss; all reported flow models use this form.

Table 10: Latent diagnostics of the grid JEPA per model seed: squared wall amplitude and total teacher-forced one-step residual at rollout steps 1 and 8, the mean wall coordinate of the one-step prediction at step one (0 = wall-free latent, 1 = true latent), the decoded wall intensity of the model's own rollout at steps 1 and 2, the fitted termination head's AUC on border and wall deaths, and walls passed by PPO inside the model (mean ± s.d. over three predictor variants and three PPO seeds).

model seed	wall amplitude 1 / 8	one-step residual 1 / 8	wall coordinate, step 1	decoded wall 1 / 2	AUC border	AUC wall	PPO inside, walls
0	0.035 / 0.124	0.145 / 0.249	0.39	0.00 / 0.00	0.96	0.65	0.29±0.34
1	0.028 / 0.082	0.130 / 0.228	0.45	0.00 / 0.00	0.96	0.65	0.17±0.22
2	0.088 / 0.193	0.156 / 0.279	0.66	0.68 / 0.02	0.88	0.76	0.33±0.24

Table 11: The grid JEPA at one, two and four times the training steps (model seed 0; the 1× fit is the ten-minute budget on an uncontended A100, the longer fits ran under GPU sharing, 1,074 and 3,802 s): steps, final prediction loss, held-out reward correlation of the fitted heads, and PPO inside the model in the three predictor variants (survival / walls passed, PPO seed 0).

budget	steps	pred. loss	held-out reward r	PPO inside: original	mean of 8	one sample
1×	8500	0.0006	0.33	0 / 0.04	3 / 0.91	2 / 0.66
2×	17000	0.0004	0.32	2 / 0.53	3 / 0.98	1 / 0.30
4×	34000	0.0003	0.31	0 / 0.05	1 / 0.29	2 / 0.60

Table 12: Latent geometry of the gate rollouts (Figure 2), averaged over 64 pre-reveal contexts. The recorded branch survives the eight steps by construction; the counterfactual branch can collide under the recorded actions, so its distances include collisions and the gap-axis rows are also given on the contexts in which both branches survive. Decoded intensities are on the shared wall pixels and, separately, in the two gap bands.

squared latent distance, rollout step	1	3	5	8
grid JEPA (latent dimension 512)				
wall present vs absent (encoded true futures)	0.035	0.041	0.071	0.124
gap top vs bottom (encoded true futures, both branches survive)	0.013	0.040	0.051	0.122
predictor's one-step residual (teacher forced, total)	0.145	0.152	0.200	0.249
one-step residual along the wall direction	0.015	0.021	0.048	0.104
one-step residual orthogonal to it	0.130	0.132	0.152	0.145
one-step prediction: mean wall coordinate (0 = no wall, 1 = true latent)	0.39	0.33	0.29	0.21
same, contexts whose wall is still solid at the first predicted frame	0.37	0.35	0.32	0.24
original rollout to the recorded future	0.14	41.71	76.23	164.00
original rollout to the no-wall future	0.13	41.69	76.28	164.59
residual-head sample to the recorded future	0.10	7.53	13.77	39.74
residual-head mean of 8 to the recorded future	0.06	10.89	20.72	27.64
flow model (latent dimension 512)				
wall present vs absent (encoded true futures)	554	413	507	344
gap top vs bottom (encoded true futures, both branches survive)	182	491	609	389
one sample: gap coordinate, $\lvert\cdot\rvert$ (1 = a real gap), both survive	0.89	1.00	1.00	0.99
mean of 8: gap coordinate, $\lvert\cdot\rvert$, both survive	0.28	0.83	0.99	0.99
decoded wall intensity on the shared wall pixels				
grid JEPA, encoded true future	1.00	1.00	1.00	1.00
grid JEPA, original rollout	0.00	0.00	0.00	0.00
flow model, one sample	1.00	1.00	1.00	0.97
flow model, mean of 8: recorded gap band	0.50	0.48	0.46	0.41
flow model, mean of 8: other gap band	0.50	0.53	0.53	0.53

Table 13: Audit of the off-budget heads on the grid JEPA latent: what they read on the $25{,}000$ real evaluation transitions split by the cause of death, what the model's own eight-step rollouts from 512 real contexts return under the dataset's actions, and what the heads-MPC planner sees at horizon 3: the mean over its 64 candidate plans against the best plan it picks.

read-out on real latents	wall deaths	border deaths	alive
transitions	610	159	24231
reward head, mean prediction	-0.05	-0.89	-0.03
termination head, mean $p(\text{done})$	0.06	0.67	0.03
termination head, AUC against alive	0.65	0.96	
imagined return, horizon 8	mean	fraction > 0	
original predictor	-0.78	1%	
residual head, one sample	-0.86	1%	
residual head, mean of 8	-0.86	0%	
planner's view, horizon 3, 64 plans	mean over plans -0.84	best plan -0.46	best plan > 0 in 5%

Table 14: Round-1 gate results with 128 planning episodes and 256 policy episodes; the planning harness's random reference passes 0.09 walls with mean length 32.2.

controller	survival (%)	length	walls passed	deaths / 100 steps
random	0±0	30.4	0.06	3.29
centre controller (true state)	0±0	31.2	0.00	3.21
LeWM (class token): goal-image MPC	0±0	32.8	0.10	3.05
LeWM (class token): heads MPC	0±0	35.5	0.09	2.82
LeWM (class token): PPO inside the JEPA	0±0	27.9	0.00	3.59
grid JEPA: goal-image MPC	0±0	32.0	0.04	3.13
grid JEPA: heads MPC	0±0	36.5	0.16	2.74
grid JEPA: PPO inside the JEPA	0±0	23.5	0.04	4.26
grid JEPA + sampling predictor: heads MPC (4 samples)	0±0	31.5	0.07	3.18
grid JEPA + sampling predictor: PPO inside	2±1	50.5	0.66	1.93
flow model: PPO inside, pixels	56±3	100.9	2.81	0.44
flow model: PPO inside, latent	98±1	126.8	3.95	0.02
PPO in the real environment, 1.5M steps	91±2	122.3	3.75	0.07
PPO in the real environment, 30 min (6.9M steps)	100±0	127.9	4.00	0.00

Table 15: Decoded wall mass in the round-1 gate diagnostics: a pixel counts as wall when every channel exceeds $200/255$; encoding a real future and rolling out a predictor test different parts of the model.

model	wall mass kept, step 1	step 4	step 8	gap in the last frame
LeWM (class token): latent of the real future, decoded	49%	30%	58%	–
LeWM (class token): mean rollout	50%	31%	59%	undecided in 73%
grid JEPA: latent of the real future, decoded	100%	100%	99%	–
grid JEPA: mean rollout	1%	0%	4%	undecided in 100%
grid JEPA + sampling predictor: sampled rollout (4 samples)	22%	34%	33%	top in 11% of samples; both sides seen in 9% of contexts
flow model: sampled rollout (4 samples)	100%	100%	97%	top in 50% of samples; both sides seen in 83% of contexts

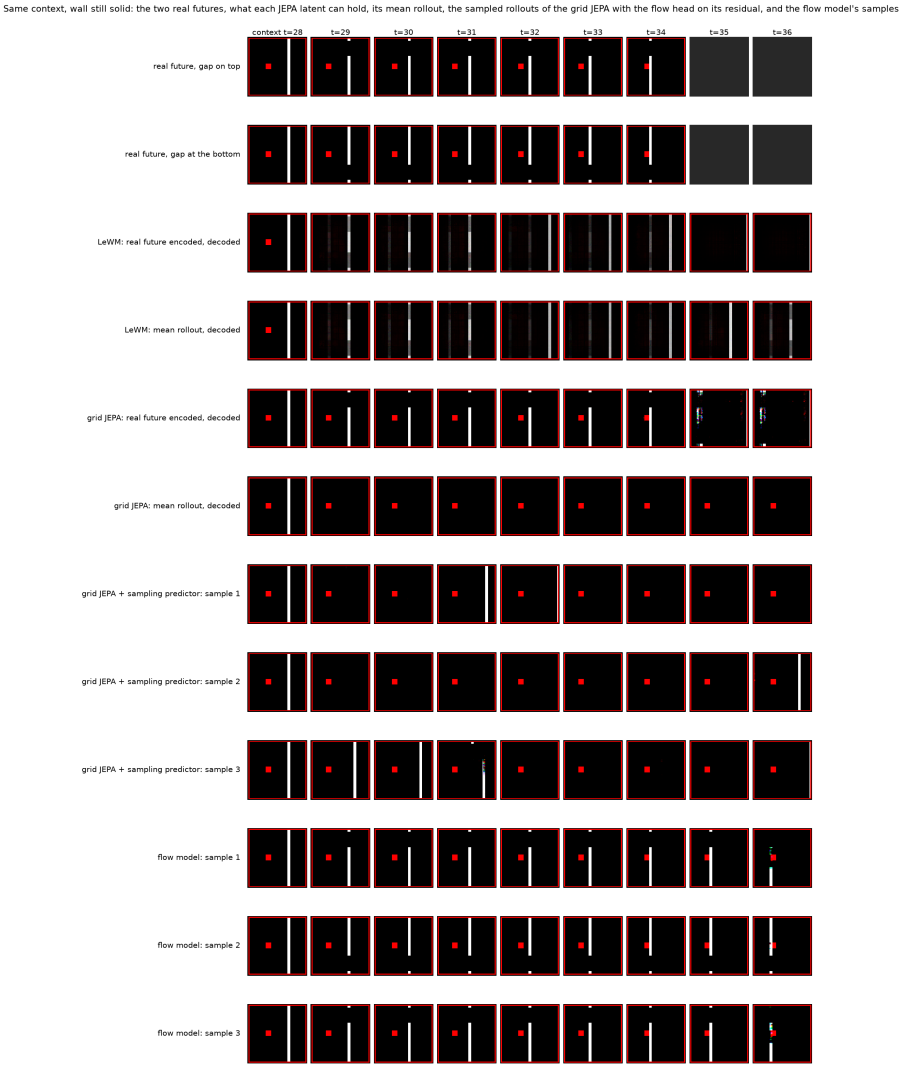


Figure 7: One pre-reveal gate context and its possible futures (round 1). Rows 1 and 2 show the real branches; rows 3 and 5 decode encoded real futures; rows 4 and 6 decode deterministic predicted rollouts; rows 7–9 show grid JEPA residual-head samples; flow samples appear below them.